



# Wibral-line PID for physical intelligence

## Results, limits and the next sensor experiment

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The immediate application is **offline sensor analysis for embodied prediction**: two cameras, with optional audio, used to predict a separately measured future physical outcome. This is an attainable first step toward robust world models and robot policies. It is not yet a demonstrated training or control improvement.

The selected functional is **Makkeh–Gutknecht–Wibral categorical shared exclusions (MGW)**, in nats, with signed atoms. The defining paper is distinct from continuous shared exclusions and from other PID definitions. Fitted bins define a categorical estimand; raw sensor information is a different object.

## The practical finding

An extra sensor's value is not its synergy alone. For two-source atoms  $R, U_1, U_2, S$ , the added Shannon information is  $U_2 + S$ . If source 1 is available with probability  $a$  and source 2 with probability  $b$ , with availability indicators  $M_1, M_2$  such that  $M_1 \perp M_2$  and  $(M_1, M_2) \perp (X_1, X_2, Y)$ , the added Bayes log-loss value is

$$\Delta = b[(1 - a)R + U_2 + aS].$$

Both predictors must see the same mask, and the complete-data policy and target law must remain fixed. Redundancy can provide backup value; unique information and synergy combine when sensors are present. The full proof includes correlated failures and data-dependent missingness in the detailed report.

For independent fair bits  $A, B, U$ , set the same target  $Y = (A, B)$  and baseline  $X_1 = A$ . Candidate noise  $U$  and useful complement  $B$  both have MGW synergy  $\ln(4/3)$ . Their added information is respectively zero and  $\ln 2$ . A duplicate  $A$  has zero synergy but supplies backup value when the first copy is unavailable.



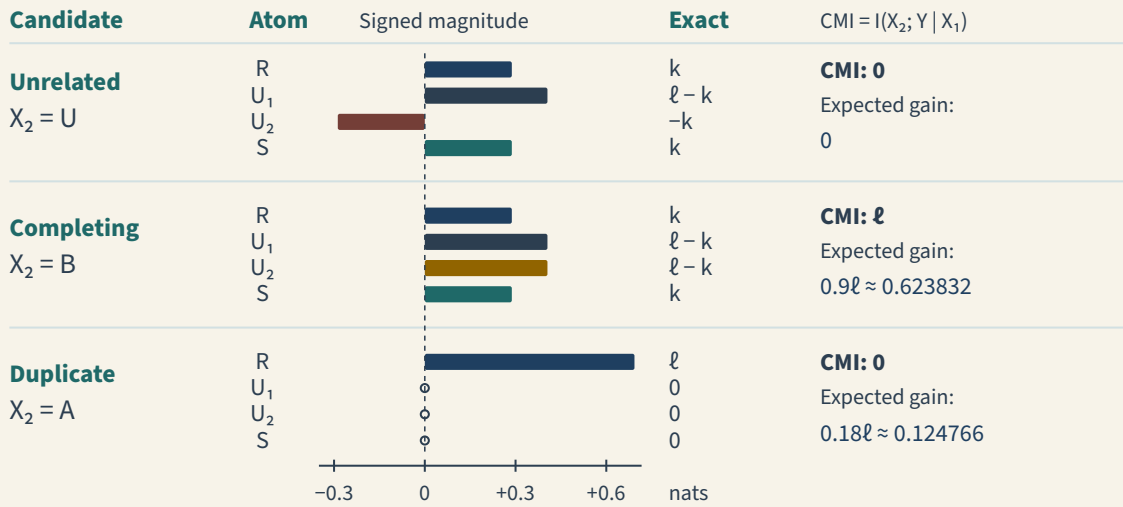


## Same synergy, different added information

Exact analytic example · signed net MGW atoms in nats

A, B, U independent fair bits; target  $Y = (A, B)$ ; baseline  $X_1 = A$ .

$\ell = \ln 2 \approx 0.693147$ ;  $k = \ln(4/3) \approx 0.287682$



U and B have the same synergy  $k$ , but their added information is 0 versus  $\ln 2$ .

Gain:  $a = 0.8$ ,  $b = 0.9$ ; observed availability masks independent of data and each other.

Bayes log-loss information gain; learned-policy benefit is unproved. Zero atoms use dots.

**Figure 1:** One target law, three candidate sensors. Signed unique information distinguishes noise from a useful complement despite equal synergy; duplication can have backup value.

There is also a useful limit: the expected Bayes log-loss reduction relative to the no-source predictor, under a fixed law and exogenous availability, depends only on subset mutual information. For two to four sources, that is 3, 7 or 15 values. Full PID's 4, 18 or 166 atoms provide a finer allocation, but no extra decision information for that exact objective. PID must earn additional cost through a useful diagnostic question or a separately justified objective.

## What exists and what it supports

| Work                                                      | Useful role                                                                 | Remaining boundary                                                                                      |
|-----------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Rust categorical MGW for 2–4 sources                      | Offline signed atom calculation with budgets and cancellation               | No completed sensor benchmark follows from library availability                                         |
| Four classical finite-PMF foundations                     | Mass/measure, integral, Gibbs and log-score results checked locally in Lean | Unconditional law only; 13 downstream conditional sensor/application targets remain open                |
| Equal-synergy and target-copy formal work                 | Reject an invalid synergy-only selection rule                               | Exact theorem premises and replay status remain in the theorem maps                                     |
| Fixed-alphabet continuity and dependence-aware law bounds | Conditional tools for estimator sensitivity                                 | A valid sampling/dependence model is still required                                                     |
| Finite-prefix mean, bias and gradient work                | Mathematical tools for a future bounded encoder experiment                  | Not an implemented robot-training method                                                                |
| Availability exposition                                   | Complete classical derivation, controls and experiment design               | Conditional prediction and availability/mask arguments remain handwritten; no scientific-priority claim |





The [mathematical results](#) guide points to standalone proofs and exact formal status. A verified mathematical intermediate step does not establish estimator calibration or physical value.

## Rust implementation and computational cost

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Rust supplies budgeted categorical MGW for two to four sources through the [averaged API](#). Event calculations are quadratic in the occupied complete states  $K$  for each lattice node. Averaged output saves retained pointwise results; row and PMF storage remain. Dense log-score validation and evaluation over  $K_Y$  target labels take  $O(K_Y)$  work and  $O(1)$  extra storage beyond the inputs.

No Rust log-score evaluator or learner is added; floating-point execution is unverified. Feature extraction, masked prediction, training and resampling add costs. Begin offline: latency and learning benefit are unmeasured. See the [full cost discussion](#).

## The ecosystem connection

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CREBAIN produces observations; NCP transports sensor records; Prisoma inspects experiments; pid-rs computes information; Galadriel can retain diagnostic evidence. Manwe supplies a separate tracking setting. The detailed report distinguishes existing consumers from proposals.

The current two-camera-plus-audio example records five images and 800 pressure samples over six 120-Hz body ticks. That proves a capture composition, not independent statistical replication. The missing step is a frozen feature table joined to future reference labels. Use one declared episode/landmark unit, preserve timing and missingness, fit transforms only on training episodes, and evaluate untouched episodes.

## How practical value will be tested

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Start offline with all eight sensor subsets. Measure predictive loss, availability, runtime and memory. Keep fixed-model outages separate from retrained subset models. Compare MI/CMI, subset task loss and established masking methods. [Masked sensor training](#) and [MAD multi-view reinforcement learning](#) are relevant baselines; no superiority to them is established here.

The detailed report derives a conservative paired-episode loss bound and states its independence and bounded-probability assumptions. It also explains why natural occlusion needs conditional laws, why more sensors can defeat finite-data estimation, and why an action-conditioned latent must not be evaluated against its own injected action.

Testing control, placement or re-identification requires explicit target, policy, association and cost contracts. Require useful task gains against matched baselines and retain negative results.

**How to cite:** Sepehr Mahmoudian, *Wibral-line PID for physical intelligence*, pid-rs technical report, 2026. Include the exact repository commit or release. Use the [citation metadata](#) for the software and cite the defining method papers separately.

